Activity 5 Report: Database Maintenance and Monitoring

COMP 018 Database Administration | Project tested: OJT Management System (Java Swing + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	OJT Management System
Owning group	[Owning group no. and name]
Repository	[Repository URL of the OJT Management System]
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a5_health.txt, a5_logs.txt, a5_maintenance.txt

Run on 6 November 2026 at 23:04 (Asia/Manila), after the application click-through of Activity 3. Results come from the saved files.

2. Health and activity

Check	Result
Version	PostgreSQL 18.6, x86_64 Windows
Database size	9006 kB
Connections	1 backend (the monitoring session). max_connections is 100
Transactions	475 commits, 22 rollbacks (the rollbacks come from the access tests)
Cache	50,375 buffer hits against 43 reads, about 99.9% hit ratio
Deadlocks and temp files	0 and 0
Autovacuum	on. No table had been vacuumed or analyzed before this run
Rows	users 5, requirements 3, student_requirements 3, one row each in students, companies, supervisors, applications, placements, attendance, evaluations; audit_log 0

Dead tuples before maintenance: student_requirements 5, users 1, attendance 1, ojt_applications 1. They come from the updates made during the click-through and are below the autovacuum thresholds.

3. Logging

Setting	Value	Comment
log_statement	none	No statement logging
log_min_duration_statement	-1	Slow queries are not logged. Recommend 500 ms
Log folder	C:/Program	Readable

	Files/PostgreSQL/18/data/log	

4. Server log findings

Time	Entry	Explanation
22:46:28	Two syntax errors	The stray ");" and the hidden character in the documented SQL (Activity 3)
22:48:18	column "employee_number" of relation "academic_supervisors" does not exist	The seed failure of the first start (Activity 3)
23:03:46	password authentication failed for user "postgres"	A mistyped password at a prompt
23:03:57 to 23:03:58	permission denied for table ..., permission denied to create role and database, must be owner of table students	The deliberate access tests of Activity 4
23:04:09	permission denied to vacuum and analyze "users"	ojt_admin trying maintenance (see section 5)
Earlier in the day	Errors and failed logins from other projects tested on the same server	Not part of this project

No unexplained errors from the application itself.

5. Maintenance performed

Action	Result
ANALYZE (all tables)	Completed. last_analyze set on all 12 tables
VACUUM (VERBOSE, ANALYZE) public.users	1 dead row version removed, 5 live rows remain, 0.00 s. n_dead_tup for users went from 1 to 0
VACUUM and ANALYZE as ojt_admin	Skipped with "permission denied to vacuum". The application roles own nothing and the project defines no maintenance role or pg_maintain grant

6. Query plan

Query: pending applications joined with students and companies. EXPLAIN (ANALYZE, BUFFERS) showed nested loops with sequential scans, planning time 4.5 ms and execution time 0.039 ms. The query returned 0 rows because the only application was already approved. With 1 row per table a sequential scan is the right plan.

The database has 24 indexes, all primary keys and unique constraints. Foreign key columns used in joins have none, for example ojt_applications.student_id and company_id, ojt_placements.student_id, company_sup_id and academic_sup_id, and evaluations.placement_id.

7. Recommendations

- Create a maintenance role with pg_maintain (PostgreSQL 17 or newer) and use it for scheduled VACUUM and ANALYZE.
- Set log_min_duration_statement to a value such as 500 ms and keep log_statement at ddl.
- Add indexes on the foreign key columns above before the tables grow.
- Add monitoring for dead tuples and long transactions, and give a monitoring role pg_monitor.

8. Feedback sent to owners

Sent on 6 November 2026: no health problems found. Recommended a maintenance role, slow-query logging and indexes on foreign keys.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own server, and every result in this report comes from the saved output files.